✅ LOCKED DECISION: Q6_K on Node D (GMKtec K15 – 48 GB DDR5)

Final Recommendation

Use Gemma-4 26B A4B Q6_K as the primary heavy model on Node D.

Why Q6_K over Q5_K_M?

Quant	Approx. RAM Usage (128k context + KV cache)	Quality vs FP16	Agentic Coding / Long Reasoning	Headroom on 48 GB DDR5	Speed Impact
Q5_K_M	~21–22 GB	~97–98%	Excellent	~22–24 GB	Slightly faster
Q6_K	**~23–24.5 GB**	**~99%+**	**Best** (noticeable on long sessions, tool chains, complex refactors)	**~20–22 GB**	Negligible slower

Key Reasons for Q6_K on K15:

- **48 GB DDR5** gives you **comfortable headroom** (~20+ GB) for:
  - Hermes Supervisor + GEPA v2 (population 12–16 + reflection)
  - Mooncake KV spillover
  - Parallel Ray tasks / swarm orchestration
  - System + OS + provenance ledger
- Q6_K is **nearly lossless** for agentic coding workflows (your primary use case). The difference becomes noticeable on long-context dev sessions, multi-step tool calls, and GEPA self-evolution.
- MoE architecture (only ~3.8–4B active params per token) keeps speed excellent even at Q6.
- Future-proof: When you add OCuLink eGPU later, Q6 scales beautifully.

Q5_K_M is acceptable only if you ever see memory pressure during extreme 200k+ context + heavy GEPA runs. For now, **Q6_K is the correct production choice.**

Node D Model Farm Topology (Locked Post-Deployment)

- **Primary Heavy Brain:** `gemma-4-26b-a4b-it-Q6_K.gguf` (Hermes Command Center / GEPA / agentic coding)
- **Fallback / Specialist** (optional, runs in parallel):
  - Qwen2.5-Coder 14B Q6 (fast code-specific)
  - Smaller Gemma E4B Q4 for quick internal tasks

Hermes Router Update (already prepared in previous patterns):

```rust
// When profile = "dev-mode" or complexity > 0.65 → Node D Q6
Backend::NodeD("gemma-4-26b-q6")
```

Immediate Next Steps (Coder-Agent Ready)

Reply with **one** of the following and I will deliver the exact files:

1. **“Give me full Nix + llama.cpp / vLLM launch command for Q6 on K15”** → including systemd service, Mooncake config, and memory tuning.
2. **“Update hermes-inference-router + GEPA for Q6 preference”**
3. **“Both – complete Node D ignition package”** (recommended)

This locks the heavy model decision and lets us move full dev work inside Machina with Hermes learning alongside you at maximum intelligence.

Status: Node D is now optimally configured for the full multi-model dev suite you’ve been building toward.

Let’s ignite it. What’s your choice?